

Short Reflection

Working on the Assignment and Grade Tracker taught me a lot about how the different concepts I learned in Python over the past 7 weeks can work together in a real project. At the beginning, I mainly understood classes, loops, and functions individually, but building this project helped me understand how they can be combined. I learned more about object-oriented programming, especially inheritance, class methods, and how objects can share information through a class. I also became more comfortable using lists, sets, loops, `try/except` (which is a really new concept), sorting, and the `datetime` module. One thing I found especially useful was learning how to handle assignments that have not been graded yet without breaking the rest of the program.

One of the first challenges I faced was trying to display assignment lists in a readable way. One of the biggest challenges I faced was getting my classes and methods to work together correctly. I made several mistakes with inheritance and `super()`, and I initially struggled with understanding why I needed to keep an assignment as an object and also understanding when to use a class method decorator. I also had difficulties with calculating averages for each subject and finding a way to calculate the highest and lowest-scoring assignments. Another challenge was input validation. I had to figure out how to prevent users from entering invalid menu options (numbers greater than 6 or even worse strings, invalid scores, incorrect dates that have already passed, incorrect date formats, or letters when the program was expecting a number). Debugging these problems took a lot of trial and error, watching youtube videos, and seeking guidance from more experienced friends, but it helped me understand my code much better.

If I had more time and more knowledge, I would improve the project by making the code more organized and reducing repeated code in my filtering functions. I would also add features such as editing and deleting assignments, allowing a user to add a grade to an assignment that was previously ungraded, and saving assignments so that they are not lost when the program closes and also undoing entries and basic assertions.. Overall, this project helped me become more confident with Python because I had to solve problems myself instead of only writing small examples.